OpenAI Drawing Extraction Pipeline

Complete Technical Documentation

Version 3.0 - Layer 1, 2, and 3 Architecture

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This document provides a comprehensive explanation of the OpenAI pipeline architecture including all stages, layers, the fallback system, quality metrics, verification systems, and CATIA integration.

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1. Executive Summary

The OpenAI Drawing Extraction Pipeline is a sophisticated multi-layer system designed to extract dimensional information from 2D mechanical engineering drawings and integrate it with CATIA 3D CAD systems for automated tolerance analysis.

The pipeline operates across THREE LAYERS with a total of 16 STAGES:

Layer 1: Drawing Extraction (8 Stages)

PDF preprocessing, AI extraction using GPT models, local OCR verification, bbox refinement, value verification, output generation, database import, and human review.

Layer 2: CATIA Parameter Integration (3 Stages)

Published parameter extraction from CATIA parts, dimension-to-parameter mapping UI, and linking relationships between 2D dimensions and 3D parameters.

Layer 3: Tolerance DMU Analysis (5 Stages)

Maximum/minimum tolerance actuation in CATIA, DMU workbench initialization, clash/clearance analysis, and result visualization.

KEY FEATURES:

- Hybrid Model Strategy: Uses GPT-5.4 Mini as primary, falls back to GPT-5.5 for quality issues
- Automatic Quality Detection: Monitors extraction quality and triggers fallback when needed
- Multi-Pass Extraction: Separates structure, dimensions, and GD&T into distinct passes
- Local OCR Verification: CRAFT + GLM-OCR validates extracted values against drawing
- Human Review Loop: Allows manual verification and dimension addition
- CATIA Integration: Automated parameter actuation and DMU clash analysis

2. Pipeline Architecture Overview

The pipeline follows a layered architecture where each layer builds upon the outputs of the previous layer. The system is designed with resilience in mind, featuring automatic fallback mechanisms and quality gates between stages.

2.1 Execution Flow

```
LAYER 1: Drawing Extraction
    |
    +-- Stage 01: PDF Clarity Preparation (image preprocessing)
    +-- Stage 02: OpenAI Extraction (GPT vision model)
    +-- Stage 03: Local BBox Refinement (CRAFT + GLM-OCR)
    +-- Stage 04: Dimension Verification (green-box counting)
    +-- Stage 05: Value Verification (OCR text matching)
    +-- Stage 06: Output Writer (JSON, annotated images)
    +-- Stage 07: SQLite Import (database storage)
    +-- Stage 08: Human Review (manual verification UI)
    |
    v (if quality checks fail -> FALLBACK to stronger model)

LAYER 2: CATIA Parameter Integration
    |
    +-- Stage 01: Published Parameter Extraction
    +-- Stage 02: Extraction Summary / JSON Output
    +-- Stage 03: Dimension Link UI
    |
    v

LAYER 3: Tolerance DMU Analysis
    |
    +-- Stage 01: Maximum Tolerance Actuation
    +-- Stage 02: DMU Mode Selection
    +-- Stage 03: Root Document Activation
    +-- Stage 04: DMU Clash Analysis
    +-- Stage 05: DMU Preview UI
```

2.2 Entry Point

The pipeline is invoked via:

```
..\.venv\Scripts\python.exe ".\openai_pipeline\5_pipeline.py" --image
".\CAS1-Suspension\3289050-1_1 3.pdf" --pdf_dpi 450
```

Default output is written to: Results/drawing_pipeline/

3. Layer 1 - Drawing Extraction (8 Stages)

Layer 1 is the core extraction layer that processes 2D engineering drawings and extracts dimensions, tolerances, GD&T symbols, notes, title blocks, and BOMs.

3.1 Stage 01: PDF Clarity Preparation

Purpose: Prepares the input drawing image for optimal OCR and AI extraction.

Operations performed:

- Grayscale conversion and contrast enhancement
- Sharpness enhancement using UnsharpMask filter
- Auto-contrast adjustment (1% cutoff)
- Resolution scaling (minimum 1600px shortest side)
- PDF rendering at configurable DPI (default: 450 DPI)
- Dynamic crop preprocessing for notes, title block, and drawing views
- Red markup box detection for user-annotated regions

Image Enhancement Parameters:

```
Contrast Enhancement: 1.25x
Sharpness Enhancement: 1.35x
UnsharpMask: radius=1.2, percent=165, threshold=2
Minimum Resolution: 1600px (shortest side)
```

3.2 Stage 02: OpenAI Extraction

Purpose: Uses OpenAI GPT vision models to extract structured data from the drawing.

Extraction Modes:

- Single-Pass: Extracts everything in one API call (faster, less precise)
- Multi-Pass (Recommended): Separates extraction into specialized passes

Multi-Pass Architecture:

```
Pass 1 - Structure Pass:
  - Document name, number, unit
  - Title block and fields
  - Bill of materials
  - Notes and note blocks
  - Drawing views (boundaries and types)

Pass 2 - Dimension Pass:
  - All visible dimensions
  - Tolerance values (+/-, limits, fits)
  - Dimension types (linear, diameter, angle)
  - View assignments

Pass 3 - Dimension Recovery Pass:
  - Finds missing dimensions
  - Focuses on small, stacked, dense clusters
  - Checks for offset/unilateral tolerances

Pass 4 - GD&T Pass:
  - Feature control frames
  - Datum identifiers
  - Geometric tolerances
```

Extraction Rules (Conservative Approach):

- Extract ONLY what is directly visible - no inference or guessing
- Every dimension must have raw_text evidence
- Tolerances are highest priority but must be visibly present
- Do not create dimensions from geometry alone
- Do not infer symmetric, repeated, or standard dimensions

3.3 Stage 03: BBox Refinement

Purpose: Refines the bounding boxes returned by OpenAI using local OCR to ensure accurate spatial positioning of extracted text.

Technology Stack:

- CRAFT (Character Region Awareness For Text detection) - text region detection
- GLM-OCR (zai-org/GLM-OCR) - text recognition
- PyMuPDF (fitz) - PDF text extraction for additional grounding

Refinement Process:

```
1. Run CRAFT text detector on preprocessed image
   -> Produces text region bounding boxes

2. For each detected region, run GLM-OCR
   -> Produces recognized text

3. For each OpenAI-extracted dimension:
   a. Normalize the raw_text (remove special chars, lowercase)
   b. Find matching CRAFT regions using similarity score
   c. If match found (score > threshold):
      - Update bbox to CRAFT detection
      - Mark annotation_source = "craft_refinement"
      - Mark annotation_ready = true
   d. If no match:
      - Keep original bbox
      - Mark annotation_source = "vlm_original"
      - May require manual review
```

Why BBox Refinement is Critical:

- GPT vision models estimate positions, local OCR measures precisely
- Accurate bboxes enable proper annotation overlay
- Required for human review UI to highlight correct regions
- Enables value verification by location matching

3.4 Stage 04: Dimension Verification

Purpose: Counts and categorizes dimensions based on their annotation readiness, producing the "green box" count used for quality assessment.

Green Box Concept:

```
A dimension is "green box" (annotation_ready = true) when:
  1. It has a valid bounding box
  2. The bbox has been verified/refined by local OCR
  3. The text value matches the detected region

Green Box Count = Number of verified, locatable dimensions
Missing Green Box = Extracted but not verified (needs review)
```

Verification Output Metrics:

- extracted_dimension_count: Total dimensions extracted by OpenAI
- green_box_dimension_count: Dimensions with verified locations
- missing_green_box_dimension_count: Dimensions needing review
- Per-view breakdown of dimension counts

3.5 Stage 05: Value Verification

Purpose: Cross-checks extracted dimension VALUES against all text found in the drawing using CRAFT/GLM-OCR. This validates that the AI did not hallucinate values.

Verification Logic:

```
For each extracted dimension:
  1. Take the raw_text value
  2. Search all CRAFT-detected text regions for matches
  3. Calculate similarity score (normalized Levenshtein ratio)
  4. Apply threshold based on text length:
     - 1-2 chars: 100% match required
     - 3-4 chars: 92% match required
     - 5-6 chars: 82% match required
     - 7+ chars: 68% match required

  5. Check if matching text bbox overlaps with dimension bbox
     - If yes: value_verified=true, location_verified=true
     - If no: value_verified=true, location_verified=false (bbox mismatch)
     - If no match found: value_verified=false (not in drawing)
```

Verification Statuses:

- verified: Both value and location confirmed
- value_correct_bbox_wrong: Value exists but in different location
- not_found_in_drawing: Value not found anywhere (possible hallucination)
- no_text: Dimension has no raw_text (invalid extraction)

3.6 Stage 06: Output Writer

Purpose: Generates all output artifacts from the extraction.

Output Files:

- JSON extraction result with all dimensions, tolerances, views
- Annotated image with colored bounding boxes
- Matplotlib overlay visualization
- Crop preprocessing debug images
- Verification metadata JSON

3.7 Stage 07: SQLite Import

Purpose: Stores extraction results in a relational database.

Database Tables:

- documents: Drawing metadata (part_number, document_id, etc.)
- dimensions: All extracted dimensions with tolerances
- views: Drawing view definitions
- catia_published_parameters: Linked CATIA parameters (Layer 2)
- catia_dimension_links: 2D-to-3D mappings (Layer 2)
- catia_dmu_results: DMU clash results (Layer 3)

3.8 Stage 08: Human Review

Purpose: Provides a UI for humans to verify, correct, and add dimensions.

Review Features:

- Visual dimension coverage preview
- Per-dimension approval/rejection
- Manual dimension entry (nominal, +tol, -tol)
- Click-to-set bounding box
- Batch actions for tolerances
- Pipeline rerun request option

4. Layer 2 - CATIA Parameter Extraction (3 Stages)

Layer 2 connects the 2D drawing dimensions to the 3D CATIA model parameters.

4.1 Stage 01: Published Parameter Extraction

Purpose: Extracts all published parameters from the CATIA part.

- Connects to running CATIA instance via COM/pycatia
- Walks product structure to find target part by part number
- Extracts Length, Angle, and other parameter types
- Stores parameters in SQLite catia_published_parameters table

4.2 Stage 02: Extraction Summary

Purpose: Generates a JSON summary of extracted CATIA parameters.

4.3 Stage 03: Dimension Link UI

Purpose: UI for manually linking 2D drawing dimensions to 3D CATIA parameters.

- Shows extracted dimensions alongside CATIA parameters
- Fuzzy name matching suggestions
- Stores links in catia_dimension_links table
- Links enable Layer 3 tolerance actuation

5. Layer 3 - Tolerance DMU Analysis (5 Stages)

Layer 3 performs automated tolerance stack-up analysis by actuating CATIA parameters to their maximum/minimum tolerance values and running DMU clash detection.

5.1 Stage 01: Maximum Tolerance Actuation

For each linked dimension:

- Calculate max value = nominal + tolerance_plus
- Calculate min value = nominal + tolerance_minus
- Set CATIA parameter to extreme value
- Update geometry

5.2 Stage 02: DMU Mode Selection

Configures DMU analysis parameters:

- Analysis type: Selection Against All
- Clearance threshold (default: 5.0mm)
- Contact and clash detection modes

5.3 Stage 03: Root Document Activation

Ensures correct CATIA document context for DMU analysis.

5.4 Stage 04: DMU Clash Analysis

Executes the clash computation and collects results:

- Clash type (contact, penetration, clearance)
- Conflict status (OK, warning, interference)
- Penetration depth values
- Affected product pairs

5.5 Stage 05: DMU Preview UI

Displays clash analysis results in a visual interface.

6. The Fallback System

The fallback system is a critical quality assurance mechanism that automatically switches to a more capable (and expensive) model when the primary model produces poor quality results.

6.1 Hybrid Model Strategy

Default Configuration:

```
Primary Model:   gpt-5.4-mini  (faster, cheaper)
Fallback Model:  gpt-5.5       (slower, more capable)
Strategy:        hybrid        (try primary, fallback if needed)
```

The pipeline first attempts extraction with GPT-5.4 Mini. If quality metrics indicate problems, it automatically reruns with GPT-5.5.

6.2 Fallback Triggers

The fallback is triggered when ANY of these conditions occur:

```
TRIGGER 1: No views extracted
  -> Indicates complete extraction failure

TRIGGER 2: Zero dimensions extracted
  -> No dimensional content found

TRIGGER 3: Extracted dimensions below expected floor
  -> Floor = max(min_threshold, view_count if views >= 2)
  -> Default min_threshold = 1

TRIGGER 4: Green-box ratio below threshold
  -> Ratio = green_box_count / extracted_count
  -> Threshold = 0.55 (55%)
  -> Only applies if extracted_count >= 4

TRIGGER 5: BBox mismatches exceed limit
  -> Value found but location wrong
  -> Default limit = 3 mismatches

TRIGGER 6: Manual review flag raised
  -> Value verification detected issues
```

6.3 Fallback Assessment Function

The assess_fallback_need() function evaluates all criteria:

```
def assess_fallback_need(
    result,
    min_extracted_dimensions=1,
    min_green_box_dimensions=1,
```

```
    min_green_box_ratio=0.55,
    max_bbox_mismatches=3,
):
    # Extracts metrics
    extracted = result["extracted_dimension_count"]
    green_box = result["dimension_count"]
    bbox_mismatches = result["bbox_mismatch_count"]
    view_count = len(result["views"])

    # Builds reasons list
    reasons = []

    if view_count == 0:
        reasons.append("no views extracted")

    if extracted == 0:
        reasons.append("no dimensions extracted")

    # ... additional checks ...

    return {
        "should_fallback": bool(reasons),
        "reasons": reasons,
        "metrics": { ... }
    }
```

6.4 Why the Fallback System Exists

1. COST OPTIMIZATION:

- GPT-5.4 Mini is significantly cheaper than GPT-5.5
- Most drawings extract well with the smaller model
- Only complex/low-quality inputs need the expensive model

2. QUALITY ASSURANCE:

- Automatic detection of poor extractions
- No manual intervention needed for common failure modes
- Ensures minimum quality standards are met

3. RESILIENCE:

- System adapts to input complexity
- Graceful degradation on difficult inputs
- Multiple chances for successful extraction

7. Quality Loss Detection

How GPT-5.4 Mini "Understands" Quality

The system does NOT rely on GPT to self-report quality. Instead, quality is measured OBJECTIVELY through post-processing verification stages.

IMPORTANT: GPT Does NOT Evaluate Its Own Output

The model simply extracts what it sees. Quality assessment is performed by independent verification stages that compare GPT output against ground truth from local OCR systems.

7.1 What "Quality" Means in This Pipeline

Quality is measured across multiple dimensions:

DIMENSION COVERAGE:

- How many dimensions were extracted vs. expected
- A drawing with 50 visible dimensions should not return only 5

BBOX ACCURACY:

- Do the extracted bounding boxes match actual text locations?
- Measured by IoU (Intersection over Union) with CRAFT detections

VALUE CORRECTNESS:

- Are the extracted values actually present in the drawing?
- Verified by text similarity matching against OCR results

TOLERANCE COMPLETENESS:

- Were tolerance-bearing dimensions captured?
- Tolerances are highest priority but easy to miss if small/faint

7.2 Quality Metrics Summary

```
METRIC                     | THRESHOLD     | MEANING
---------------------------|---------------|-------------------------------
view_count                 | > 0           | At least one drawing view found
extracted_dimension_count  | >= 1          | At least one dimension extracted
green_box_ratio            | >= 0.55       | 55%+ dimensions have valid bbox
bbox_mismatch_count        | <= 3          | Max 3 location mismatches allowed
manual_review_required     | false         | No verification failures
```

7.3 The Verification Chain

Quality is established through a chain of independent verifiers:

```
GPT Extraction
      |
      v
[CRAFT Text Detection] -- Finds all text regions in drawing
      |
      v
[GLM-OCR Recognition] -- Reads text from detected regions
      |
      v
[BBox Refinement] -- Matches GPT boxes to CRAFT boxes
      |
      v
[Value Verification] -- Checks if values exist in drawing
      |
      v
[Quality Assessment] -- Computes metrics, decides fallback
      |
      v
If quality < threshold --> FALLBACK to GPT-5.5
If quality >= threshold --> Continue to Output
```

Each verifier operates independently of GPT. If CRAFT detects 100 text regions but GPT only found 10 dimensions, that discrepancy indicates quality loss.

8. BBox Refinement Deep Dive

Why is bbox refinement necessary?

Vision Language Models (VLMs) like GPT are excellent at READING text but only ESTIMATE positions. Their bounding boxes are approximations, not precise measurements.

```
GPT Bbox (estimated):  [120, 340, 180, 360]  -- rough guess
CRAFT Bbox (measured): [125, 342, 175, 358]  -- pixel-accurate

The CRAFT bbox comes from actual edge detection of character pixels.
The GPT bbox comes from the model's spatial reasoning about the image.
```

8.1 The CRAFT Algorithm

CRAFT (Character Region Awareness For Text detection) is a deep learning model that detects text regions by predicting character and inter-character affinity maps.

- Input: Preprocessed drawing image
- Output: List of text region bounding boxes
- Model: craft_mlt_25k.pth (multi-language trained)
- Thresholds: text_threshold=0.7, low_text=0.4

8.2 The Matching Process

```
1. Normalize both texts:
   - Remove special characters
   - Lowercase
   - Replace diameter symbols
   - Remove whitespace

2. Calculate similarity score:
   - SequenceMatcher ratio
   - Score = number of matching chars / total chars

3. Apply distance weighting:
   - Prefer matches closer to original bbox
   - Center distance penalty for far matches

4. Select best match:
   - Above similarity threshold
   - Closest to original location
```

9. Dimension Verification Explained

Dimension verification determines which extracted dimensions are "annotation ready" - meaning they have accurate positions and can be displayed on the drawing.

9.1 The Green Box System

Green Box = Verified Dimension

A dimension with annotation_ready=true has a confirmed bounding box that can be reliably used to draw annotation overlays on the drawing image.

Annotation Sources:

- craft_refinement: Bbox refined by CRAFT+GLM-OCR (highest confidence)
- pdf_text_grounding: Bbox from PDF embedded text (high confidence)
- vlm_original: Original GPT bbox, unverified (lower confidence)

9.2 Verification Statistics

```
Output metrics from dimension_verifier.py:

{
  "status": "verified" | "partial",
  "extracted_dimension_count": 45,
  "green_box_dimension_count": 38,
  "missing_green_box_dimension_count": 7,
  "views": [
    {
      "view_id": "MAIN",
      "extracted_dimension_count": 30,
      "green_box_dimension_count": 26,
      "missing_green_box_dimension_count": 4
    },
    ...
  ]
}
```

10. Value Verification System

Value verification answers a critical question: Is the extracted dimension value actually present in the drawing, or did the AI hallucinate it?

10.1 The Verification Process

```
For dimension with raw_text = "45.5 +/-0.1":

1. Run CRAFT + GLM-OCR on entire drawing
   -> Produces list of all text with positions

2. Normalize target text: "455pm01" (simplified)

3. Search all detected texts for match:
   - "45.5+/-0.1" at [200,300,250,320] -> Score: 0.98
   - "45.5" at [400,100,430,115] -> Score: 0.72
   - "455" at [600,500,625,515] -> Score: 0.68

4. Best match: "45.5+/-0.1" with score 0.98

5. Check location overlap:
   - Dimension bbox: [195, 295, 255, 325]
   - Match bbox: [200, 300, 250, 320]
   - IoU = 0.85 -> Location verified!

Result: value_verified=true, location_verified=true
```

10.2 Verification Statuses

- verified: Value found AND location matches (ideal)
- value_correct_bbox_wrong: Value found but in wrong place
- not_found_in_drawing: Value does not exist in drawing (hallucination!)
- no_text: Dimension was extracted without raw_text (invalid)

10.3 Similarity Thresholds

```
Text Length | Required Match Score | Rationale
------------|---------------------|--------------------------
1-2 chars   | 100%                | "5" must match exactly
3-4 chars   | 92%                 | "12.5" allows minor OCR error
5-6 chars   | 82%                 | "+/-0.1" has more room
7+ chars    | 68%                 | Long text tolerates more noise
```

11. Quality Metrics Summary

Complete list of quality metrics tracked by the pipeline:

```
EXTRACTION METRICS:
  - view_count: Number of drawing views extracted
  - extracted_dimension_count: Total dimensions from GPT
  - tolerance_bearing_count: Dimensions with tolerances
  - gdt_count: Feature control frames extracted
  - note_count: Notes and specifications extracted

VERIFICATION METRICS:
  - green_box_dimension_count: Dimensions with verified bbox
  - green_box_ratio: green_box / extracted (target > 0.55)
  - bbox_mismatch_count: Values found but locations wrong
  - not_found_count: Values not in drawing (hallucinations)
  - no_text_count: Dimensions missing raw_text

QUALITY GATES:
  - views > 0: At least one view required
  - extracted > floor: Minimum dimension threshold
  - green_ratio >= 0.55: Majority verified
  - bbox_mismatches <= 3: Limited location errors

FALLBACK TRIGGER:
  Any quality gate failure -> Switch to GPT-5.5
```

Appendix A: Command Reference

```
BASIC USAGE:
python openai_pipeline/5_pipeline.py --image <path>

COMMON OPTIONS:
--image <path>              Input drawing (PDF or image)
--pdf_dpi <number>          PDF render DPI (default: 450)
--pdf_page <number>         PDF page number (default: 1)
--model <name>              Fallback model (default: gpt-5.5)
--primary_model <name>      Primary model (default: gpt-5.4-mini)
--model_strategy <mode>     hybrid|direct (default: hybrid)
--result_folder <path>      Output directory

EXTRACTION OPTIONS:
--extraction_mode <mode>    single|multi (default: multi)
--crop_preprocessing_mode   skip|dynamic|red_boxes

VERIFICATION OPTIONS:
--value_verification_mode   skip|local (default: local)
--bbox_refinement_mode      skip|local (default: local)

LAYER 2/3 OPTIONS:
--catia_published_parameters_mode  skip|extract
--catia_dimension_link_mode        skip|link
--catia_layer3_mode               skip|run|ask

EXAMPLE:
python openai_pipeline/5_pipeline.py \
  --image "./drawings/part_123.pdf" \
  --pdf_dpi 500 \
  --model_strategy hybrid \
  --extraction_mode multi \
  --value_verification_mode local
```

Appendix B: Configuration Options

B.1 Fallback Thresholds

```
--fallback_min_extracted_dimensions  (default: 1)
--fallback_min_green_box_dimensions  (default: 1)
--fallback_min_green_box_ratio       (default: 0.55)
--fallback_max_bbox_mismatches       (default: 3)
```

B.2 OCR Configuration

```
--bbox_refiner_glm_model    (default: zai-org/GLM-OCR)
--bbox_refiner_max_new_tokens  (default: 512)
```

B.3 CATIA Configuration

```
--catia_root_document       Path to CATProduct/CATPart
--catia_published_part_number  Target part number
--dmu_clearance_mm          DMU clearance threshold (default: 5.0)
--catia_layer3_output_root  Layer 3 output folder
```